

6th Grade Review

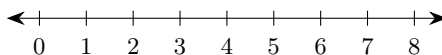
Name: _____

Date: _____

Directions: No calculator. Write decimal answers as decimals. Write fraction answers in simplest terms. Show your work on scrap paper.

A. Number Lines

1. Plot the point 4 on the number line below.



B. Order of Operations

1. $8 + 3 \times 4 =$
2. $(21 - 4) \times 3 =$
3. $2^4 + 1 =$
4. $(5 + 1)^2 =$
5. $3(2^2) =$
6. $(3 \times 2)^2 =$

C. Decimals, Fractions, and Percents

1. In 82.731, the tenths digit is _____.
2. $314.7 + 29.58 =$ _____ (decimal)
3. $81.5 - 7.94 =$ _____ (decimal)
4. $0.32 =$ _____ %
5. $65\% =$ _____ as a decimal
6. $0.8 =$ _____ as a fraction
7. 15% of 80 = _____

D. Facts and Decimal Multiply/Divide

1. $9 \times 6 =$
2. $63 \div 9 =$
3. $2.4 \times 1.5 =$ _____ (decimal)
4. $5.6 \div 0.8 =$ _____

E. Ratios, Fractions, and Proportions

1. Which is greater: $\frac{5}{8}$ or 0.61?
2. $\frac{4}{7} = \frac{?}{21}$
3. 4 snacks cost \$20. 7 snacks cost _____ dollars.
4. $\frac{5}{6} = \frac{x}{18}$

F. Vocabulary, Expressions, and Equations

1. In $7m$, the number 7 is the _____.
2. In $n + 8 = 12$, n is the _____.
3. To undo $+8$, use the inverse operation _____.
4. To undo $\div 5$, use the inverse operation _____.
5. Combine like terms: $6y - 2y =$
6. If $n = 4$, then $5n - 3 =$

G. Distributive Property

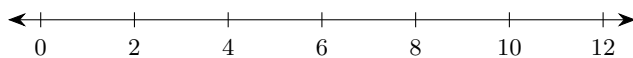
1. Expand: $3(y + 6) =$

H. Equations, Inequalities, and Graphing

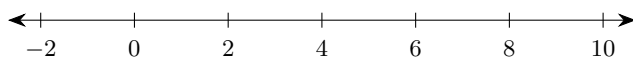
1. Solve: $x + 7 = 18$
2. Solve: $\frac{x}{4} = 6$
3. Solve: $x + 3 > 8$
4. Solve: $x - 4 < 2$

5. Write an equivalent way to express $6 < y$.

6. Graph $6 < y$ on the number line below.



7. Graph $x > 2$ on the number line below.



I. Geometry

1. Rectangle: length 9, width 4. Area =

2. Rectangle: length 11, width 3. Perimeter =

3. Rectangle: area is 42 square units and width is 6. Length =

J. Units and Quantities

1. 3 hours = _____ minutes

2. $\frac{3}{5}$ of 35 =

3. 210 miles $\div$ 3 hours = _____ miles per hour

4. 72 inches = _____ yards

5. 150 miles $\div$ 5 gallons = _____ miles per gallon

K. Expressions, Factors, and Fraction Structure

1. Is $8x + 1$ a linear expression?

2. Prime factorization of 90 =

3. Find the GCF of 27 and 36.

4. Simplify $\frac{27}{36}$.

5. Convert $2\frac{2}{3}$ to an improper fraction.

6. Which is greater: $1\frac{3}{4}$ or $(1)(\frac{3}{4})$?

7. In $\frac{m}{2}$, the coefficient of m is _____.

8. In $\frac{q}{5}$, the coefficient of q is _____.
9. Rewrite using the commutative property: $8 + 5 =$
10. Rewrite using the commutative property: $7 \times a =$
11. Rewrite using the associative property: $(b + 3) + 9 =$

L. Distributive Property and Factoring

1. Factor: $8x + 16 =$

M. Decimal Computation

1. $428.5 + 37.26 =$ _____ (decimal)
2. $700.3 - 284.67 =$ _____ (decimal)
3. $18.4 \times 12 =$ _____ (decimal)
4. $168.8 \div 8 =$ _____ (decimal)
5. $4.2 \times 1.5 =$ _____ (decimal)
6. $7.2 \div 0.8 =$ _____

N. Fraction Operations and Properties

1. $\frac{1}{2} + \frac{1}{4} =$ _____ (simplest fraction)
2. $\frac{5}{6} - \frac{1}{3} =$ _____ (simplest fraction)
3. $\frac{5}{8} \times \frac{4}{15} =$ _____ (simplest fraction)
4. $\frac{7}{8} \div \frac{1}{4} =$ _____ (simplest fraction)
5. Rewrite using the commutative property: $9 \times b =$
6. Rewrite using the associative property: $(x + 5) + 8 =$
7. Is subtraction associative? Compare $(12 - 4) - 3$ and $12 - (4 - 3)$.